

**DIPLOMA EXAMINATION IN
ENGINEERING/TECHNOLOGY/MANAGEMENT/COMMERCIAL
PRACTICE**

ADVANCED AUTOMOBILE ENGINEERING

[Time: 3 hours]

(Maximum Marks: 100)

Marks

PART-A

(Maximum Marks: 10)

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. What is D MPFI?
2. What are the main areas of pollution in an IC engines?
3. List effects of pollutants.
4. List any two types of seat belts?
5. Define GPS. (5x2=10)

PART-B

(Maximum Marks: 30)

II. Answer any five questions. Each question carries 6 marks.

1. Discuss the advantages of MPFI.
2. Explain how diesel smoke controlled.
3. Discuss the factors affecting diesel smoke.
4. Explain EGR system.
5. Discuss the working of an electric mirror.
6. Explain about infotainment system in automobiles.
7. Describe vehicle immobilizer. (5x6=30)

PART-C

(Maximum marks: 60)

(Answer one full question from each unit. Each question carries 15 marks)

UNIT-I

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| III. | a. Explain the working of CRDI system with a neat sketch.. | 8 |
| | b. What are the advantages of petrol injection system over carburettor system? | 7 |

OR

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| IV. | a. Explain working of any two actuators used in MPFI system. | 8 |
| | b. Differentiate D MPFI and L MPFI . | 7 |

UNIT-II

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| V. | a. What is meant by a crankcase blowby? How it can be controlled? | 8 |
| | b. Explain PCV system in automobile with neat sketch.. | |

7

OR

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| VI. | a. Explain evaporative emission control system using charcoal canister. | 8 |
| | b. Explain BHARATH STAGE emission control standards. | 7 |

UNIT-III

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| VII. | a. Explain the working of electric seat in an automobile. | 8 |
| | b. Explain the working of automobile air condition system. | 7 |

OR

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| VIII. | a. Explain the working of automatic climate control used in automobiles. | 8 |
| | b. Explain cruise control system | 7 |

UNIT-IV

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| IX. | a. Explain the working of ABS with the help of a sketch. | 8 |
| | b. Explain the working of an airbag system. | 7 |

OR

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| X. | a. Describe automatic traction control system system. | 8 |
| | b. Explain EBD | 7 |